EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND
INFORMATION TECHNOLOGY

Week 6 Practice Assignment – Quarter 2, 2025-2026
Method: LAB 5 Practice 5
Course Name: Advanced Databases **Course Code:** CSW 436
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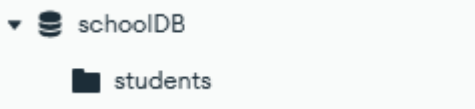
Important Instructions –

- Implement each question using MongoDB.
- find(), update(), delete().
- Use $group, $count, $avg, $max and other NoSQL method for query processing and optimization.
- Document all question Output/results properly by capturing the screenshots of the output/results and NoSQL code for every question in a Word / PDF document and save it.
- After completing all questions, upload the document to Moodle.

Topic: Key-value databases and Document databases

1. Create Database & Collection
   - Create a database named schoolDB
   - Create a collection called students



2. Insert Documents
   - **one document using insertOne()**

db.students.insertOne({insert stuff})

```
> db.students.insertOne({
```

   - **multiple documents using insertMany()**

db.students.insertOne([{ insert stuff }])

```
> db.students.insertMany([
```

   - **Insert 5 student documents with:**
     - **name**

- age
- department
- marks
- email

```
> db.students.insertOne({
    name: "John Doe",
    age: 21,
    department: "Computer Science",
    marks: 85,
    email: "john.doe@example.com"
  })
< {
    acknowledged: true,
    insertedId: ObjectId('698171d3611c0f76931fd695')
  }
```

```
> db.students.insertMany([
    {
      name: "Typhon",
      age: 22,
      department: "Computer Science",
      marks: 92,
      email: "ty.phon123@example.com"
    },
    {
      name: "zhuang fangyi",
      age: 20,
      department: "Mechanical",
      marks: 100,
      email: "zhuang.fangyi@example.com"
    },
    {
      name: "Pogranichnik",
      age: 23,
      department: "IT",
      marks: 88,
      email: "pog.pogger@example.com"
    },
```

```
    {
      name: "Lifeng",
      age: 19,
      department: "Computer Science",
      marks: 35,
      email: "lifeng.gen@example.com"
    }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6981728f611c0f76931fd696'),
      '1': ObjectId('6981728f611c0f76931fd697'),
      '2': ObjectId('6981728f611c0f76931fd698'),
      '3': ObjectId('6981728f611c0f76931fd699')
    }
  }
```

3. **Find Data**
   - **Write queries to:**
     - **Display all students**
       db.students.find({})

```
// find data
> db.students.find({})
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 85,
    email: 'john.doe@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 92,
    email: 'ty.phon123@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
```

```
  _id: ObjectId('6981728f611c0f76931fd697'),
  name: 'zhuang fangyi',
  age: 20,
  department: 'Mechanical',
  marks: 100,
  email: 'zhuang.fangyi@example.com'
}
{
  _id: ObjectId('6981728f611c0f76931fd698'),
  name: 'Pogranichnik',
  age: 23,
  department: 'IT',
  marks: 88,
  email: 'pog.pogger@example.com'
}
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 35,
  email: 'lifeng.gen@example.com'
}
```

- **Display only name and marks**
  db.students.find({}, { name: 1, marks: 1, _id: 0 })

```
> db.students.find({}, { name: 1, marks: 1, _id: 0 })
< {
    name: 'John Doe',
    marks: 85
  }
  {
    name: 'Typhon',
    marks: 92
  }
  {
    name: 'zhuang fangyi',
    marks: 100
  }
  {
    name: 'Pogranichnik',
    marks: 88
  }
  {
    name: 'Lifeng',
    marks: 35
  }
```

- **Find student with age > 20**

db.students.find({ age: { $gt: 20 } })

```
> //find student with age>20
> db.students.find({ age: { $gt: 20 } })
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 85,
    email: 'john.doe@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 92,
    email: 'ty.phon123@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
```

```
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 88,
    email: 'pog.pogger@example.com'
  }
  {
    _id: ObjectId('698172d5611c0f76931fd69a'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 85,
    email: 'john.doe@example.com'
  }
```

- **Find students from "Computer Science" department**

db.students.find({ department: "Computer Science" })

```
> db.students.find({ department: "Computer Science" })
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 85,
    email: 'john.doe@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 92,
    email: 'ty.phon123@example.com'
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 35,
  email: 'lifeng.gen@example.com'
}
```

- **Find one student using email**

db.students.findOne({ email: "alice.smith@example.com" })

```
> db.students.findOne({ email: "zhuang.fangyi@example.com" })
< {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 100,
    email: 'zhuang.fangyi@example.com'
  }
```

4. **Update Data**
   - **Write queries to:**
     - **Update marks of one student**

db.students.updateOne({ name: "Lifeng" },{ $set: { marks: 50 } })

```
> db.students.updateOne(
    { name: "Lifeng" },
    { $set: { marks: 50 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

Before:

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 35,
  email: 'lifeng.gen@example.com'
}
```

After:

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 50,
  email: 'lifeng.gen@example.com'
}
```

- **Increase all students' marks by 5**

db.students.updateMany({},{ $inc: { marks: 5 } })

```
> db.students.updateMany({},{ $inc: { marks: 5 } })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 6,
    modifiedCount: 6,
    upsertedCount: 0
  }
```

Before:

```
// find data
> db.students.find({})
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 85,
    email: 'john.doe@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 92,
    email: 'ty.phon123@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
```

```
  _id: ObjectId('6981728f611c0f76931fd697'),
  name: 'zhuang fangyi',
  age: 20,
  department: 'Mechanical',
  marks: 100,
  email: 'zhuang.fangyi@example.com'
}
{
  _id: ObjectId('6981728f611c0f76931fd698'),
  name: 'Pogranichnik',
  age: 23,
  department: 'IT',
  marks: 88,
  email: 'pog.pogger@example.com'
}
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 35,
  email: 'lifeng.gen@example.com'
}
```

After:

```
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 90,
    email: 'john.doe@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com'
```

```
    marks: 105,
    email: 'zhuang.fangyi@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com'
  }
```

- **Add new field status: "active" to all students**

db.students.updateMany({},{ $set: { status: "active" } })

```
> db.students.updateMany({},{ $set: { status: "active" } })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 6,
    modifiedCount: 6,
    upsertedCount: 0
  }
```

```
> db.students.find({})
< {
    _id: ObjectId('698171d3611c0f76931fd695'),
    name: 'John Doe',
    age: 21,
    department: 'Computer Science',
    marks: 90,
    email: 'john.doe@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
```

```
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active'
```

5. **Delete Data**

- **Write queries to:**
  - **Delete one student by name**

db.students.deleteOne({ name: "John Doe" })

```
> db.students.deleteOne({ name: "John Doe" })
< {
    acknowledged: true,
    deletedCount: 1
  }
```

```
> db.students.find({})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd697'),
  name: 'zhuang fangyi',
  age: 20,
  department: 'Mechanical',
  marks: 105,
  email: 'zhuang.fangyi@example.com',
  status: 'active'
}
{
  _id: ObjectId('6981728f611c0f76931fd698'),
  name: 'Pogranichnik',
  age: 23,
  department: 'IT',
  marks: 93,
  email: 'pog.pogger@example.com',
  status: 'active'
}
```

```
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active'
  }
```

- **Delete all students with marks < 40**

db.students.deleteMany({ marks: { $lt: 40 } })

-Insert 2 student lower than 40

```
> db.students.insertMany([
    {
      name: "randomGuy1",
      age: 22,
      department: "Computer Science",
      marks: 30,
      email: "random.guy1@example.com"
    },
    {
      name: "RandomGuy2",
      age: 20,
      department: "Mechanical",
      marks: 10,
      email: "random.guy2@example.com"
    }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('698421dc7aceb55dd05c1276'),
      '1': ObjectId('698421dc7aceb55dd05c1277')
    }
  }
```

```
> db.students.deleteMany({ marks: { $lt: 40 } })
< {
    acknowledged: true,
    deletedCount: 2
  }
```

```
> db.students.find({})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com',
    status: 'active'
  }
```

```
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active'
  }
```

-> Successfully delete all student that score is <40

6. Filtering with Operators

• Write queries using:

- $gt, $lt
- $in
- $and, $or
- $ne

Tasks:

- **Students with marks between 60 and 90**

  // Use $gt and $lt

-Insert 2 student with point 60 and 68

```
> db.students.insertMany([
    {
      name: "Mifu",
      age: 22,
      department: "Mechanical",
      marks: 60,
      email: "mi.fufu@example.com"
    },
    {
      name: "Vladimir",
      age: 20,
      department: "Mechanical",
      marks: 68,
      email: "bob.johnson@example.com"
    }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69845836dab13107b9cc3e77'),
      '1': ObjectId('69845836dab13107b9cc3e78')
    }
  }
```

db.students.find({marks: { $gt: 60, $lt: 90 }})

```
> db.students.find({marks: { $gt: 60, $lt: 90 }})
< {
    _id: ObjectId('69845836dab13107b9cc3e78'),
    name: 'Vladimir',
    age: 20,
    department: 'Mechanical',
    marks: 68,
    email: 'bob.johnson@example.com'
  }
```

- **Students not from "Mechanical"**

// Use $ne

db.students.find({department: { $ne: "Mechanical" }})

```
> db.students.find({department: { $ne: "Mechanical" }})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 55,
  email: 'lifeng.gen@example.com',
  status: 'active'
}
```

○ **Students from "CS" or "IT"**

// Use $in

db.students.find({department: { $in: ["Computer Science", "IT"] }})

```
> db.students.find({department: { $in: ["Computer Science", "IT"] }})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
```

```
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active'
  }
```

// Use $and

db.students.find({$and: [{ age: { $gte: 20 } },{ marks: { $gte: 80 } }]})

```
  }
> db.students.find({$and: [{ age: { $gte: 20 } },{ marks: { $gte: 80 } }]})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com',
    status: 'active'
  }
```

```
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
```

//Using $or operator

db.students.find({$or: [{ department: "Computer Science" },{ department: "IT" }]})

```
> db.students.find({$or: [{ department: "Computer Science" },{ department: "IT" }]})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 55,
  email: 'lifeng.gen@example.com',
  status: 'active'
}
```

7. Sorting & Limiting
   - Sort students by marks descending
     db.students.find({}).sort({ marks: -1 })

```
> db.students.find({}).sort({ marks: -1 })
< {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
```

```
  _id: ObjectId('6981728f611c0f76931fd698'),
  name: 'Pogranichnik',
  age: 23,
  department: 'IT',
  marks: 93,
  email: 'pog.pogger@example.com',
  status: 'active'
}
{
  _id: ObjectId('69845836dab13107b9cc3e78'),
  name: 'Vladimir',
  age: 20,
  department: 'Mechanical',
  marks: 68,
  email: 'bob.johnson@example.com'
}
{
  _id: ObjectId('69845836dab13107b9cc3e77'),
  name: 'Mifu',
  age: 22,
  department: 'Mechanical',
  marks: 60,
  email: 'mi.fufu@example.com'
}
```

```
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active'
  }
```

- Show top 3 students
  db.students.find({}).sort({ marks: -1 }).limit(3)

```
> db.students.find({}).sort({ marks: -1 }).limit(3)
< {
    _id: ObjectId('6981728f611c0f76931fd697'),
    name: 'zhuang fangyi',
    age: 20,
    department: 'Mechanical',
    marks: 105,
    email: 'zhuang.fangyi@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
```

- Skip first 2 and show next 3

db.students.find({}).sort({ marks: -1 }).skip(2).limit(3)

```
> db.students.find({}).sort({ marks: -1 }).skip(2).limit(3)
< {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active'
  }
  {
    _id: ObjectId('69845836dab13107b9cc3e78'),
    name: 'Vladimir',
    age: 20,
    department: 'Mechanical',
    marks: 68,
    email: 'bob.johnson@example.com'
  }
```

```
  }
  {
    _id: ObjectId('69845836dab13107b9cc3e77'),
    name: 'Mifu',
    age: 22,
    department: 'Mechanical',
    marks: 60,
    email: 'mi.fufu@example.com'
  }
```

8. Arrays Practice
   - Add field:
   - subjects: ["DBMS", "OS", "CN"]

db.students.updateMany({},{$set:{subjects:["DBMS","OS","CN"]}})

```
> db.students.updateMany({},{$set:{subjects:["DBMS","OS","CN"]}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 6,
    modifiedCount: 6,
    upsertedCount: 0
  }
```

Tasks:

- Find students who have "DBMS"

db.students.find({subjects: "DBMS"})

```
> db.students.find({
    subjects: "DBMS"
  })
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ]
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd697'),
  name: 'zhuang fangyi',
  age: 20,
  department: 'Mechanical',
  marks: 105,
  email: 'zhuang.fangyi@example.com',
  status: 'active',
  subjects: [
    'DBMS',
    'OS',
    'CN'
  ]
}
{
  name: 'Pogranichnik',
  age: 23,
  department: 'IT',
  marks: 93,
  email: 'pog.pogger@example.com',
  status: 'active',
  subjects: [
    'DBMS',
    'OS',
    'CN'
  ]
}
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 55,
  email: 'lifeng.gen@example.com',
  status: 'active',
  subjects: [
    'DBMS',
    'OS',
    'CN'
  ]
```

```
    age: 22,
    department: 'Mechanical',
    marks: 60,
    email: 'mi.fufu@example.com',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ]
  }
  {
    _id: ObjectId('69845836dab13107b9cc3e78'),
    name: 'Vladimir',
    age: 20,
    department: 'Mechanical',
    marks: 68,
    email: 'bob.johnson@example.com',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ]
  }
```

- Add new subject

db.students.updateOne({ name: "Mifu" },{ $push: { subjects: "Endfield" } })

```
> db.students.updateOne({ name: "Mifu" },{ $push: { subjects: "Endfield" } })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
> db.students.find({subjects: "Endfield"})
< {
    _id: ObjectId('69845836dab13107b9cc3e77'),
    name: 'Mifu',
    age: 22,
    department: 'Mechanical',
    marks: 60,
    email: 'mi.fufu@example.com',
    subjects: [
      'DBMS',
      'OS',
      'CN',
      'Endfield'
    ]
  }
```

- Remove one subject

db.students.updateOne({ name: "Mifu" },{ $pull: { subjects: "CN" } })

```
> db.students.updateOne({ name: "Mifu" },{ $pull: { subjects: "CN" } })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
< {
    _id: ObjectId('69845836dab13107b9cc3e77'),
    name: 'Mifu',
    age: 22,
    department: 'Mechanical',
    marks: 60,
    email: 'mi.fufu@example.com',
    subjects: [
      'DBMS',
      'OS',
      'Endfield'
    ]
  }
```

9. Embedded Documents

• Add:

address: {
city: "Ho chi Minh",
pincode: 234569
}

db.students.updateMany({},{$set:{address:{city: "Ho Chi Minh",pincode:234569}}})

```
< already on db schoolDB
> db.students.updateMany({},{$set: {address: {city: "Ho Chi Minh",pincode: 234569}}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 6,
    modifiedCount: 6,
    upsertedCount: 0
  }
schoolDB>
```

Tasks:

- Find students from Delhi
  db.students.find({"address.city": "Delhi"})

```
> db.students.find({"address.city": "Delhi"})
<
```

- Insert a student live at delhi:

```
> db.students.insertOne({
    name: "Arclight",
    age: 22,
    department: "Computer Science",
    marks: 85,
    email: "arclight@example.com",
    subjects: ["DBMS", "OS", "Endfield"],
    address: {
      city: "Delhi"
    }
  })
< {
    acknowledged: true,
    insertedId: ObjectId('6986dbcc766c5c03c1b57031')
  }
```

```
> db.students.find({"address.city": "Delhi"})
```

```
{
  _id: ObjectId('6986dbcc766c5c03c1b57031'),
  name: 'Arclight',
  age: 22,
  department: 'Computer Science',
  marks: 85,
  email: 'arclight@example.com',
  subjects: [
    'DBMS',
    'OS',
    'Endfield'
  ],
  address: {
    city: 'Delhi'
  }
}
```

- Update pincode

db.students.updateMany({"address.city": "Ho Chi Minh"},{$set:{ "address.pincode":700000}})

```
> db.students.updateMany({"address.city": "Ho Chi Minh"},{$set:{ "address.pincode":700000}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 6,
    modifiedCount: 6,
    upsertedCount: 0
  }
```

```
> db.students.find({})
< {
    _id: ObjectId('6981728f611c0f76931fd696'),
    name: 'Typhon',
    age: 22,
    department: 'Computer Science',
    marks: 97,
    email: 'ty.phon123@example.com',
    status: 'active',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ],
    address: {
      city: 'Ho Chi Minh',
      pincode: 700000
    }
  }
```

```
    address: {
      city: 'Ho Chi Minh',
      pincode: 700000
    }
  }
  {
    _id: ObjectId('6981728f611c0f76931fd698'),
    name: 'Pogranichnik',
    age: 23,
    department: 'IT',
    marks: 93,
    email: 'pog.pogger@example.com',
    status: 'active',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ],
    address: {
      city: 'Ho Chi Minh',
      pincode: 700000
    }
  }
  {
    _id: ObjectId('6981728f611c0f76931fd699'),
```

- Add street field
  db.students.updateOne({ name: "Lifeng" },{ $set: { "address.street": "Wuling" }})

```
> db.students.updateOne({ name: "Lifeng" },{ $set: { "address.street": "Wuling" }})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
{
  _id: ObjectId('6981728f611c0f76931fd699'),
  name: 'Lifeng',
  age: 19,
  department: 'Computer Science',
  marks: 55,
  email: 'lifeng.gen@example.com',
  status: 'active',
  subjects: [
    'DBMS',
    'OS',
    'CN'
  ],
  address: {
    city: 'Ho Chi Minh',
    pincode: 700000,
    street: 'Wuling'
  }
}
```

10.Aggregation Framework Use aggregate() to:

- Count total students
  db.students.aggregate([{ $count: "totalStudents" }])

```
> db.students.aggregate([{ $count: "totalStudents" }])
< {
    totalStudents: 8
  }
```

- Find average marks
  db.students.aggregate([{$group: {_id: null,averageMarks: { $avg: "$marks" }}}])

```
> db.students.aggregate([{$group: {_id: null,averageMarks: { $avg: "$marks" }}}])
< {
    _id: null,
    averageMarks: 81
  }
```

- Find highest marks

db.students.aggregate([{$group:{_id:null,highestMarks:{$max:"$marks"}}}])

```
> db.students.aggregate([{$group: {_id: null,highestMarks: { $max: "$marks" }}}])
< {
    _id: null,
    highestMarks: 105
  }
schoolDB>
```

- Group students by department and count them
  db.students.aggregate([{$group:{_id:"$department",count:{ $count:{}}}}])

```
> db.students.aggregate([{$group:{_id:"$department",count:{ $count:{}}}}])
< {
    _id: 'IT',
    count: 2
  }
  {
    _id: 'Computer Science',
    count: 3
  }
  {
    _id: 'Mechanical',
    count: 3
  }
```

- Find department with highest average marks
  db.students.aggregate([{$group: {_id: "$department",averageMarks: { $avg: "$marks"}}},{$sort: { averageMarks: -1 }},{$limit: 1}])

```
> db.students.aggregate([{$group: {_id: "$department",averageMarks: { $avg: "$marks"}}},{$sort: { averageMarks: -1 }},{$limit
< {
    _id: 'IT',
    averageMarks: 89
  }
```

Activate Windows
Go to Settings to activate Windows

11.Indexing

- Create index on email
  db.students.createIndex({email:1})

```
> db.students.createIndex({email:1})
< email_1
schoolDB>
```

- Create compound index on department + marks
  db.students.createIndex({department: 1, marks: -1})

```
> db.students.createIndex({department: 1, marks: -1})
< department_1_marks_-1
```

  View all index:
  db.students.getIndexes()

```
< department_1_marks_-1
> db.students.getIndexes()
< [
    { v: 2, key: { _id: 1 }, name: '_id_' },
    { v: 2, key: { email: 1 }, name: 'email_1' },
    {
      v: 2,
      key: { department: 1, marks: -1 },
      name: 'department_1_marks_-1'
    }
  ]
```

- Explain why indexing is useful

- Performance:
+Index speed up query operations
+Without index: MongoDB scans every column and row
+With index: MongoDB uses B-tree structure

-Useful for queries that filter on multiple fields

12.Relationships
Create:
- courses
- enrolments

Tasks:
- Insert courses

```
> db.courses.insertMany([
    {
      courseId: "CS101",
      courseName: "Intro to programmi
      credits: 3
    },
    {
      courseId: "CS102",
      courseName: "Data Structure",
      credits: 4
    },
    {
      courseId: "CS103",
      courseName: "Discreet math",
      credits: 3
    },
    {
      courseId: "IT101",
      courseName: "Web Development",
      credits: 3
    }
  ])
< {
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6986e3ff766c5c03c1b57032'),
      '1': ObjectId('6986e3ff766c5c03c1b57033'),
      '2': ObjectId('6986e3ff766c5c03c1b57034'),
      '3': ObjectId('6986e3ff766c5c03c1b57035')
    }
  }
schoolDB>
```

- Enrol students

- get student id

```
db.students.find({}, {  _id: 1, name: 1 })
```

- insert enrollment

```
schoolDB> db.enrollments.insertMany([
            {
              studentId: ObjectId("69845836dab13107b9cc3e77"),
              courseId: "CS101",
              semester: "Fall 2025"
            },
            {
              studentId: ObjectId("6981728f611c0f76931fd698"),
              courseId: "CS102",
              semester: "Fall 2025"
            }
          ])
```

```
> db.enrollments.insertMany([
    {
      studentId: ObjectId("69845836dab13107b9cc3e77"),
      courseId: "CS101",
      semester: "Fall 2025"
    },
    {
      studentId: ObjectId("6981728f611c0f76931fd698"),
      courseId: "CS102",
      semester: "Fall 2025"
    }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6986e4b1766c5c03c1b57036'),
      '1': ObjectId('6986e4b1766c5c03c1b57037')
    }
  }
```

- Use $lookup to join students with courses

```
db.enrollments.aggregate([
  {
    $lookup: {
      from: "students",
      localField: "studentId",
      foreignField: "_id",
      as: "studentDetails"
    }
  },
  {
    $lookup: {
      from: "courses",
      localField: "courseId",
      foreignField: "courseId",
      as: "courseDetails"
    }
  },
```

```
  {
    $project: {
      studentName: "$studentDetails.name",
      courseName: "$courseDetails.courseName",
      credits: "$courseDetails.credits",
      semester: 1
    }
  }
```

```
< {
    _id: ObjectId('6986e4b1766c5c03c1b57036'),
    semester: 'Fall 2025',
    studentName: 'Mifu',
    courseName: 'Intro to programming',
    credits: 3
  }
  {
    _id: ObjectId('6986e4b1766c5c03c1b57037'),
    semester: 'Fall 2025',
    studentName: 'Pogranichnik',
    courseName: 'Data Structure',
    credits: 4
  }
```

13. Build Student Result Management System

Students must:

- Insert 20 records

```
> db.students.insertMany([
    {
      name: "Nguyen Van A",
      age: 21,
      department: "Computer
      marks: 85,
      email: "nguyenvana@ex
      status: "enrolled",
      subjects: ["DBMS", "O
      address: { city: "Ho
    },
    {
      name: "Tran Thi B",
      age: 22,
      department: "IT",
      marks: 92,
      email: "tranthib@exam
      status: "enrolled",
      subjects: ["Web Dev",
      address: { city: "Ho
    },
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6986e8db766c5c03c1b57038'),
      '1': ObjectId('6986e8db766c5c03c1b57039'),
      '2': ObjectId('6986e8db766c5c03c1b5703a'),
      '3': ObjectId('6986e8db766c5c03c1b5703b'),
      '4': ObjectId('6986e8db766c5c03c1b5703c'),
      '5': ObjectId('6986e8db766c5c03c1b5703d'),
      '6': ObjectId('6986e8db766c5c03c1b5703e'),
      '7': ObjectId('6986e8db766c5c03c1b5703f'),
      '8': ObjectId('6986e8db766c5c03c1b57040'),
      '9': ObjectId('6986e8db766c5c03c1b57041'),
      '10': ObjectId('6986e8db766c5c03c1b57042'),
      '11': ObjectId('6986e8db766c5c03c1b57043'),
      '12': ObjectId('6986e8db766c5c03c1b57044'),
      '13': ObjectId('6986e8db766c5c03c1b57045'),
      '14': ObjectId('6986e8db766c5c03c1b57046'),
      '15': ObjectId('6986e8db766c5c03c1b57047'),
      '16': ObjectId('6986e8db766c5c03c1b57048'),
      '17': ObjectId('6986e8db766c5c03c1b57049'),
      '18': ObjectId('6986e8db766c5c03c1b5704a'),
      '19': ObjectId('6986e8db766c5c03c1b5704b')
    }
  }
```

- Query failed students
  db.students.find({marks:{ $lt: 50}})

```
> db.students.find({marks:{ $lt: 50}})
< {
    _id: ObjectId('6986e8db766c5c03c1b5703b'),
    name: 'Pham Thi D',
    age: 23,
    department: 'Mechanical',
    marks: 35,
    email: 'phamthid@example.com',
    status: 'enrolled',
    subjects: [
      'Thermodynamics',
      'Mechanics'
    ],
    address: {
      city: 'Da Nang',
      pincode: 550000,
      street: 'Bach Dang'
    }
  }
  {
    _id: ObjectId('6986e8db766c5c03c1b5703f'),
    name: 'Bui Thi H',
    age: 24,
    department: 'Mechanical',
```

```
    name: 'Bui Thi H',
    age: 24,
    department: 'Mechanical',
    marks: 38,
    email: 'buithih@example.com',
    status: 'enrolled',
    subjects: [
      'Fluid Mechanics',
      'Heat Transfer'
    ],
    address: {
      city: 'Can Tho',
      pincode: 900000,
      street: 'Hai Ba Trung'
    }
  }
  {
    _id: ObjectId('6986e8db766c5c03c1b57045'),
    name: 'Duong Thi O',
    age: 19,
    department: 'Computer Science',
    marks: 45,
    email: 'duongthio@example.com',
    status: 'enrolled',
```

- Top 5 scorers

db.students.find({},{ name: 1, marks: 1, department: 1,_id: 0}).sort({ marks: -1 }).limit(5)

```
> db.students.find({},{ name: 1, marks: 1, department: 1,_id: 0}).sort({ marks: -1 }).limit(5)
< {
    name: 'Do Thi K',
    department: 'IT',
    marks: 82
  }
  {
    name: 'Le Van C',
    department: 'Computer Science',
    marks: 78
  }
  {
    name: 'Ha Van T',
    department: 'Computer Science',
    marks: 74
  }
  {
    name: 'Dang Van G',
    department: 'IT',
    marks: 72
  }
  {
    name: 'Vladimir',
```

- Department wise average

db.students.aggregate([{$group: {_id: "$department",averageMarks: { $avg: "$marks" },totalStudents: { $sum: 1 },maxMarks: { $max: "$marks" },minMarks: { $min: "$marks" }}},{$sort: { averageMarks: -1 }}])

```
> db.students.aggregate([{$group: {_id: "$depart
< {
    _id: 'IT',
    averageMarks: 87.2,
    totalStudents: 10,
    maxMarks: 93,
    minMarks: 72
  }
  {
    _id: 'Computer Science',
    averageMarks: 78.81818181818181,
    totalStudents: 11,
    maxMarks: 97,
    minMarks: 45
  }
  {
    _id: 'Mechanical',
    averageMarks: 60.42857142857143,
    totalStudents: 7,
    maxMarks: 105,
    minMarks: 35
  }
```

- Update results

// give +5 for student who have <50 point

db.students.updateMany({ marks: { $lt: 50 } },{ $inc:{ marks: 5}})

```
> db.students.updateMany({ marks: { $lt: 50 } },{ $inc:{ marks: 5}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 2,
    modifiedCount: 2,
    upsertedCount: 0
  }
```

- Delete graduated students

// update for student with point >85 to graduate

db.students.updateMany({ marks:{ $gte: 85}},{ $set:{status:"graduated" }})

```
> // update for student with point >85 to graduate
  db.students.updateMany({ marks:{ $gte: 85}},{ $set:{status:"graduated" }})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 15,
    modifiedCount: 11,
    upsertedCount: 0
  }
```

// delete student graduated
db.students.deleteMany({status: "graduated"})

```
> db.students.deleteMany({status: "graduated"})
< {
    acknowledged: true,
    deletedCount: 15
  }
```

- find(), update(), delete().

//find student who pass, point must >=50
db.students.find({marks: { $gte:50}})

```
> db.students.find({marks: { $gte:50}})
< {
    _id: ObjectId('6981728f611c0f76931fd699'),
    name: 'Lifeng',
    age: 19,
    department: 'Computer Science',
    marks: 55,
    email: 'lifeng.gen@example.com',
    status: 'active',
    subjects: [
      'DBMS',
      'OS',
      'CN'
    ],
    address: {
      city: 'Ho Chi Minh',
      pincode: 700000,
      street: 'Wuling'
    }
```

```
  {
    _id: ObjectId('69845836dab13107b9cc3e77'),
    name: 'Mifu',
    age: 22,
    department: 'Mechanical',
    marks: 60,
    email: 'mi.fufu@example.com',
    subjects: [
      'DBMS',
      'OS',
      'Endfield'
    ],
    address: {
      city: 'Ho Chi Minh',
      pincode: 700000
    }
  }
  {
    _id: ObjectId('69845836dab13107b9cc3e78'),
    name: 'Vladimir',
```

```
    department: 'Computer Science',
    marks: 78,
    email: 'levanc@example.com',
    status: 'enrolled',
    subjects: [
      'DBMS',
      'AI',
      'ML'
    ],
    address: {
      city: 'Hanoi',
      pincode: 100000,
      street: 'Trang Tien'
    }
  }
  {
    _id: ObjectId('6986e8db766c5c03c1b5703e'),
    name: 'Dang Van G',
    age: 19,
    department: 'IT',
    marks: 72,
    email: 'dangvang@example.com',
    status: 'enrolled',
    subjects: [
```

- Use $group, $count, $avg, $max,

- group

db.students.aggregate([{$group: {_id: "$department",totalStudents:{$sum:1}}}])

```
> db.students.aggregate([{$group: {_id: "$department",totalStudents:{$sum:1}}}])
< {
    _id: 'IT',
    totalStudents: 2
  }
  {
    _id: 'Computer Science',
    totalStudents: 5
  }
  {
    _id: 'Mechanical',
    totalStudents: 6
  }
```

-count
db.students.aggregate([{ $count: "totalStudents" }])

```
> db.students.aggregate([{ $count: "totalStudents" }])
< {
    totalStudents: 13
  }
```

- avg
db.students.aggregate([{$group:{_id: null,averageMarks:{$avg:"$marks"}}}])

```
> db.students.aggregate([{$group:{_id: null,averageMarks:{$avg:"$marks"}}}])
< {
    _id: null,
    averageMarks: 62.76923076923077
  }
schoolDB>
```

-max
db.students.aggregate([{$group:{_id: null,highestMarks:{$max:"$marks"}}}])

```
> db.students.aggregate([{$group:{_id: null,highestMarks:{$max:"$marks"}}}])
< {
    _id: null,
    highestMarks: 82
  }
```

EIU